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TrafficFlowBench Challenge

Residual Reconstruction, Analogue Forecasting and Exact Demand Estimation for Freeway Traffic

Final Report

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Abstract

TrafficFlowBench asks for four coupled answers on ten synthetic but physically calibrated freeway corridors: reconstruct masked speed and flow, forecast which links will be queued over the next 30 minutes, keep the reconstruction consistent with the fundamental diagram and vehicle conservation, and recover path-level demand from link counts. This report describes the pipeline behind our two final submissions. State is reconstructed by gradient-boosted trees that learn the *residual* of same-day temporal interpolation from 92 features of the masked data. The predictions are reconciled with a predicted density, corrected on the free-flow branch by spatial covariance and low-rank factor models, averaged over three independently sampled fits, and recentred per link on short temporal gaps where a bias test shows the same-day interpolant to be unbiased. Queues are forecast from exactly twelve history slots: onsets by a joint-mask analogue that maximizes empirical expected intersection over union across complete training outcomes, and ongoing queues by a first-crossing classifier combined with a trajectory-analogue library through an expected-IoU decoder. Demand is the exact non-negative least-squares solution on each split's own counts and prior. Because the four tasks are scored on disjoint rows, the composite score is exactly additive, and we use our evaluated submissions as a controlled ablation that identifies the score of each component. State reconstruction and onset forecasting carry most of the improvement over the reference baselines. Every state change selected on held-out training months improved the evaluated score, whereas only half of the queue changes did. We also document more than thirty rejected approaches, including Godunov, Newell and diagram-inversion reconstructions, all of which lost to temporal interpolation on held-out months.

Keywords: traffic state estimation; kinematic wave theory; queue forecasting; analogue forecasting; gradient boosting; origin-destination estimation; reproducibility.

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1 Introduction

A traffic management centre does four things in sequence. It fills the gaps in its detector feed, it asks where congestion will be in half an hour, it checks that its picture of the road is physically possible, and it estimates where the traffic is coming from and going to. TrafficFlowBench [1] turns that loop into a single benchmark. Ten directional freeway panels with five-minute resolution and eleven months of records are scored by

$$S_{\text{total}} = 0.35\, S_{\text{state}} + 0.30\, S_{\text{queue}} + 0.15\, S_{\text{physics}} + 0.20\, S_{\text{ODME}}, \tag{1}$$

where each component lies in $[0, 1]$. The tasks interlock by design. Physics is not submitted at all: it is read off the reconstructed state. A reconstruction can match every detector to within noise and still create or destroy vehicles between neighbouring links, a failure that squared error does not detect. Queue forecasts begin from the same congested region that the reconstruction must get right, and demand estimation needs counts that are consistent with the flows.

The data are synthetic. A traffic-flow model whose parameters were measured from real freeway detector data on the same corridors generated every split, so the records obey traffic physics and carry realistic measurement error, but no external source contains the answers. The validation month (March 2031) and the private month (April 2031) are generated independently, with five and seven incidents respectively, so a method tuned to one month learns that month's incidents rather than the traffic.

1.1 Contributions

- **A residual reconstruction model** that learns the error of same-day temporal interpolation from 92 features built only from the masked layer (Section 5). It is reconciled with a predicted density, corrected on the free-flow branch by spatial covariance and low-rank factor models, averaged over three independently sampled fits, and recentred per link on short gaps where a two-stage bias test shows the same-day interpolant to be unbiased.
- **Queue forecasting under a strict twelve-slot information contract** (Section 7): forward-filled persistence that retains dropped sensors, a joint-mask onset analogue that chooses the Bayes action for intersection over union across complete training outcomes [2], and an ongoing-queue decoder that combines a first-crossing classifier with a trajectory-analogue library [3].
- **Exact split-specific demand estimation** by non-negative least squares [4], which reproduces the organizers' reference solve (Section 8).
- **The evaluation as a measurement instrument** (Section 9). The four tasks are scored on disjoint rows, so Equation (1) is exactly additive. Anchored on the organizers' baseline scores, submissions that change a single task identify the score of each component.
- **A record of what failed** (Sections 6 and 10), including Godunov [5, 6], Newell [7] and diagram-inversion reconstructions that lost to temporal interpolation on held-out months, and more than thirty further variants rejected by pre-registered acceptance tests.
- **Reproducible release engineering** (Section 12). The final reconstruction submission regenerates all 6,980,503 rows from the released data and saved weights with a single command.

1.2 Summary of findings

1. *Learn the residual, not the field.* A tree ensemble trained on the residual of same-day temporal interpolation roughly halves speed error at every mask rate and every gap length on a held-out month (Section 5).
2. *Physics is a scoring constraint here, not a better predictor.* The released diagram is satisfied almost exactly by any accurate field (S_{FD} between 0.984 and 0.994 on every panel). Godunov, steady-flux, diagram-inversion and Newell reconstructions all lost on held-out months (Section 6).
3. *Most queue skill comes from retaining information.* Forward-filling dropped sensors raises ongoing IoU by 0.135 over the organizers' persistence baseline, and choosing whole onset masks by expected IoU raises onset IoU by 0.124 over a per-link classifier (Section 7).
4. *The open problem is selection, not representation.* On the 40 published ongoing windows the analogue's candidate set contains a mask scoring 0.934 IoU, while its rule selects one scoring 0.836. No frozen selection rule we tested closed that gap (Section 7.6).
5. *Demand is solved by solving the right system.* Exact NNLS on each split's own counts raises the implied S_{ODME} from 0.836 to 0.994; every prior-modelling alternative was worse (Section 8).
6. *State proxies transfer; queue proxies are noisy.* All six state changes selected on held-out training months improved the evaluated score, whereas only five of ten queue changes did. Queue-window sampling alone has a standard deviation of about 0.0126 in S_{total} (Section 9).

1.3 The two final submissions

Two submissions were selected for final scoring (Table 1). TF3 is the cleanest end-to-end artefact: it is regenerated from the raw release by one command. ES4 extends it with an ongoing-queue classifier refitted on all nine training months and a per-link speed-bias correction on gaps of one to six slots. The two files contain the same 6,980,503 rows and differ in 5,825,667 speed values and 247 ongoing-queue labels; onset labels, flows and path flows are identical.

Table 1. The two final submissions and their validation (public) scores. No private score was available at the time of writing.

Submission	Content	Validation score
TF3	Three-member residual reconstruction, analogue queues, exact demand	0.86991
ES4	TF3 with all-month ongoing classifier and short-gap bias correction	0.87187

Only the evaluation platform returns validation scores. Every other figure in this report is a *local proxy*: an error measured on held-out training months whose labels are released, on visible cells of the evaluation months that we masked ourselves, or on the 80 published training queue windows. Proxy values are marked with † where they could be confused with validation scores, and Section 9.3 shows how often the queue proxies failed to transfer.

1.4 Organization

Section 2 describes the benchmark and the properties of its design that shaped our method. Section 3 states the information boundaries and the experimental protocol. Section 4 presents the system architecture, and Sections 5 to 8 treat the four tasks. Section 9 analyses the validation evidence, Section 10 lists negative results and Section 11 discusses error, limitations and future work. Section 12 documents reproduction.

2 The benchmark

2.1 Corridors, splits and network

The release covers five corridor families in two districts, each with two directions (Table 2). Every panel ships mainline speed, flow and occupancy at five-minute resolution, on- and off-ramp counts, link geometry, per-link triangular fundamental-diagram parameters, an LWR topology file, a path set and a weak path-flow prior. The training split runs from 1 June 2030 to 28 February 2031 (273 days) and ships an unmasked copy of every observation. Validation (March 2031, 31 days) and private (April 2031, 30 days) ship only the masked layer.

Table 2. The ten directional panels, computed from the released network files. Paths equal $Z(Z-1)/2$ for Z zones: one path per upstream-to-downstream zone pair. Task 2 excludes the two D12 I-405 panels.

Panel	Links	Length (km)	Detector links	Mean lanes	On / off ramps	Zones	Paths
D7_I10_E	146	55.6	101	4.23	62 / 54	93	4,278
D7_I10_W	107	49.1	107	4.17	60 / 55	101	5,050
D7_I210_E	297	81.3	75	4.04	56 / 42	73	2,628
D7_I210_W	81	44.5	81	3.99	62 / 45	82	3,321
D7_I405_N	352	67.2	96	4.46	61 / 42	96	4,560
D7_I405_S	109	42.0	109	4.35	62 / 51	97	4,656
D12_I5_N	258	72.6	93	4.71	63 / 49	93	4,278
D12_I5_S	100	45.0	100	4.67	53 / 54	101	5,050
D12_I405_N	89	45.3	50	5.05	31 / 23	43	903
D12_I405_S	44	32.7	44	5.02	27 / 20	36	630
Total	1,583	535.2	856		537 / 435		35,354

Figure 1 shows the released diagrams. Jam density is exactly 100 veh/km/lane on every link; free-flow speeds lie between 105 and 128 km/h, per-lane capacities mostly between 1,450 and 2,750 veh/h, and the implied backward wave speed w is about 23 km/h in District 7 and 26 km/h in District 12. Links are short: the shortest is 10 m and many are a few hundred metres, so a five-minute bin is long compared with the free-flow travel time of a link.

2.2 Masking and eligibility

A cell is *eligible* when its detector reported for at least 75 % of the interval and the required values are present; about 76 % of cells qualify, and only eligible cells are scored in any task. Task 1 removes eligible cells under three regimes, R1, R2 and R3, at 20, 30 and 50 % of cells. Each calendar day belongs to exactly one regime, so no cell hidden in one view is visible in another. At a blanked cell every measured channel is removed: speed, flow, occupancy and the occupancy-derived density. The quality flag (`pct_observed`) stays, so the target cells are known. The Task 2 horizon and the 60 minutes after it are also blanked so that the horizon cannot be interpolated across.

2.3 Tasks and scoring

Task 1, state. For each regime r, speed and per-lane flow errors are scored separately and blended,

$$S_{\text{state}}(r) = 0.54\ \max\!\Big(0,\ 1 - \frac{\text{RMSE}_v(r)}{25\ \text{km/h}}\Big) + 0.46\ \max\!\Big(0,\ 1 - \frac{\text{RMSE}_{q/\text{lane}}(r)}{600\ \text{veh/h}}\Big),$$
$$S_{\text{state}} = \tfrac{1}{3}\textstyle\sum_{r\in\{R1,R2,R3\}} S_{\text{state}}(r). \tag{2}$$

The equal weight on regimes means the 50 % mask counts as much as the 20 % mask.

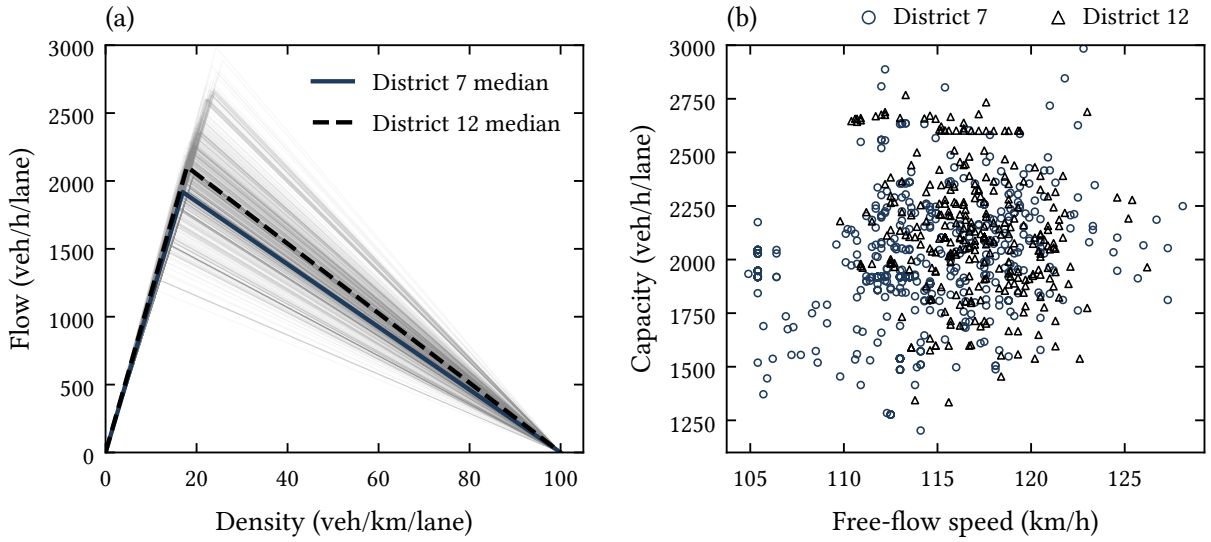


Figure 1. The physics the evaluator uses. Left: the per-lane triangular diagram of every link (grey) and the median diagram of each district. Right: free-flow speed against per-lane capacity. Task 3 checks the submitted speed and flow against these diagrams and against vehicle conservation.

Task 2, queue. A link is *queued* when its latent speed is at most $0.6\,v_f$. Each window provides the history up to the forecast origin T and asks for the queue field $\hat{Y} \in \{0,1\}^{L\times 6}$ at $T{+}5, \ldots, T{+}30$ minutes. A window is scored by space-time intersection over union [8],

$$\mathrm{IoU}(\hat{Y}, Y) = \frac{|\hat{Y} \wedge Y|}{|\hat{Y} \vee Y|}, \tag{3}$$

and S_{queue} is the mean over windows, five *onset* and five *ongoing* per panel and split. Onset windows have no queued link anywhere in the history and at least one in the horizon; ongoing windows have a queue at T. Because the two conditions have equal counts, $0.30\,S_{\text{queue}} = 0.15\,\mathrm{IoU}_{\text{onset}} + 0.15\,\mathrm{IoU}_{\text{ongoing}}$.

Task 3, physics. There are no Task 3 rows. The evaluator forms $k = q/v$ and $N = kL$ from the submitted state and scores

$$S_{\text{physics}} = \tfrac{1}{3}\,S_{\text{FD}} + \tfrac{2}{3}\,S_{\text{LWR}}, \qquad N_{t+\Delta t} - N_t \stackrel{!}{=} \Delta t\,\big(q_{\text{in}} + q_{\text{on}} - q_{\text{out}} - q_{\text{off}}\big), \tag{4}$$

where S_{FD} measures the normalized distance between the submitted flow and the diagram flow at the same density [9, 10], and S_{LWR} the size of the conservation residual over link-to-link transitions with the released ramps. All ten corridors use coverage mode A (0.756 ramp coverage).

Task 4, demand. With path-link incidence A, released counts c and prior b, $S_{\text{ODME}} = 0.45\,S_{\text{od}} + 0.25\,S_{\text{link}} + 0.15\,S_{\text{dev}} + 0.15\,S_{\text{attr}}$ compares the submitted path flows f to a withheld reference, Af to c, the distance moved from b to the reference's, and the destination-attraction pattern.

Aggregation. Each component is computed per panel, the two directions of a corridor are averaged, and the five corridor families are averaged with equal weight. A task with a missing or incomplete row set scores zero.

2.4 Anatomy of a submission

A submission has 6,980,503 rows: 6,735,795 *state* rows (speed and flow for every Task 1 target cell of validation and private), 174,000 *queue* rows (8 panels × 10 windows × 1,450 links × 6 horizons, summed over two splits, half onset and half ongoing) and 70,708 *ODME* rows (35,354 paths × 2 splits). This arithmetic matters for the analysis in Section 9: the three blocks are disjoint, so each task's score depends only on its own rows.

2.5 Five properties of the design that shaped our method

P1. Physics without lookup. The generator obeys LWR-type physics with calibrated noise, but no external record contains the answers. Methods must model the release, not join against outside data.

P2. Onsets are placed at the end of the horizon. The onset selector takes the earliest qualifying origin, one whose hour of history has no queue and whose horizon does, with selected origins at least 360 minutes apart. Simulating that rule on the training split reproduced 39 of the 40 published onset origins. A consequence is that the first queued cell almost always appears at $T{+}30$, the last horizon step. The release notes that onsets are visible in advance: an hour before a queue forms the road runs at about 1.28 times its queue threshold, against 1.54 times where none follows, and carries about 6,912 veh/h against 4,620.

P3. Ongoing windows are those where persistence fails. A window is kept only if copying the current queue forward scores an IoU of at most 0.9 on it. Trivial persistence is therefore weaker than on random congested windows by construction.

P4. Queue scores are noisy. The organizers report a per-window IoU standard deviation of 0.3752 on the persistence baseline; with five windows per condition and panel this is about 0.0126 of S_{total} from window sampling alone. Validation and private queue scores can differ by that much with no change in skill.

P5. Physics cannot be scored locally. The conservation check needs the organizer's boundary flows, which are withheld. Estimating them from the submitted flows swamps the signal: the conservation signal is about 0.75 % of the vehicles on a link and that estimate is off by about 0.22 %, so a perfect answer on D12_I5_N scores 0.3305 locally and 0.9604 with the organizer flows.

Table 3. Organizer reference scores on the validation split, from the baseline table of the benchmark repository. These values anchor the additive decomposition of Section 9; our submission of the same baselines scored 0.55343.

Component	Naive baseline	Baseline score	Perfect answer
S_{state}	weekday × time-of-day historical mean	0.6929	1.0000
S_{queue}	persistence of the last history slot	0.3017	1.0000
S_{physics}	(inherits the state baseline)	0.3549	0.9636
S_{ODME}	NNLS against TRAIN counts	0.8359	1.0000
S_{total}		0.5534	0.9945

3 Information boundaries and research protocol

3.1 What each task may read

Tasks 1, 3 and 4 are offline: the rules allow any released observation of the evaluation split, at any time, including times after the target cell. Task 2 is online: a forecast for origin T may use released data stamped at or before T. We adopted a stricter contract for Task 2 and kept it in every selected file (Table 4 and Figure 2). A forecast reads *exactly* the twelve released slots $T-60, \dots, T-5$ of its own window, static network data, and models or analogue libraries fitted on the training split. It does not read slot T (which the current rules allow), other days or windows of the evaluation month, or the offline state reconstruction. The organizer's own persistence baseline was rewritten to the same twelve slots during the competition.

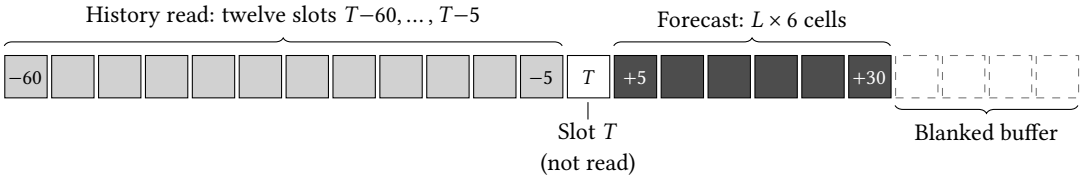


Figure 2. Task 2 information contract used in both final submissions. Each window reads only its own twelve history slots (light grey), together with static network data and models fitted on the training split. Slot T, other days and windows of the evaluation month, data after T and the offline Task 1 reconstruction are not read.

Table 4. Information sets. "Allowed" is the competition rule; "used" is what the selected files read.

Task	Allowed by the rules	Used by TF3 and ES4
1 State	Any released observation of the split, any time; train in full	Masked layer of the same split (target channels absent by construction); TRAIN-fitted trees, profiles, factor and covariance models
2 Queue	Released data stamped $\leq T$, including earlier days of the split	Exactly $T-60 \dots T-5$ of the window; static network; TRAIN-fitted classifier and TRAIN analogue library
3 Physics	No rows; read from Task 1	Not applicable
4 ODME	Split's released counts, prior and incidence	Same; each split solved on its own counts and prior, never TRAIN counts

One early diagnostic file (`P-slotT`) set ongoing queues from the state at slot T in order to measure what that slot adds. It was never a candidate and is not part of either final submission. Section 7.3 shows that a forward-fill of the twelfth history slot scores higher than that diagnostic.

3.2 A hierarchy of proxies

Hidden labels are held only by the evaluation platform, so every decision used one of four local instruments, each with a known weakness.

(i) **Held-out training months.** Models are fitted on June–November and scored on December, January and February with the official masks and the official scorers for Tasks 1 and 4. This is the cleanest instrument, but training months share one generator draw: they are not new incident schedules.

(ii) **The 80 published training queue windows** (40 onset, 40 ongoing), all dated 1–10 June 2030, scored with latent labels reconstructed from the unmasked layer. They are few and repeatedly inspected.

(iii) **Visible-cell pseudo-events in March and April.** Extra masks or pseudo-windows are drawn on cells the release shows, with odd and even dates split into development and confirmation halves. The targets are noisy observations, not organizer latent labels.

(iv) **A local conservation proxy** that replaces the withheld boundary flux by the observed accumulation change. We verified that our implementation equals the official Task 3 scorer given the same flux file, but the proxy sits near the exact-answer ceiling and is not the official physics score.

3.3 Pre-registered gates

From 3 October onward each candidate change was registered before its result was seen, with a fixed development rule, a single confirmation and no rescue. The state ensemble (TS3, Section 5.6) illustrates the template:

- *Development* on December and January with models fitted on June–November: the joint proxy $0.35\,S_{\text{state}} + 0.15\,S_{\text{phys}}^{\text{proxy}}$ must rise by at least 0.0001, S_{state} must not fall, at least seven of ten panels must improve and no panel may lose more than 0.001.
- *One confirmation* on February with the same rule and at least eight improving panels. A failed arm is closed; no member, weight or panel subset is re-selected afterwards.
- *Final refit* of the frozen rule on all nine training months, with unchanged seeds and settings.

Queue changes used the analogous structure on visible March/April events (odd dates for development, even dates for confirmation) and on the published June windows, with date-clustered bootstrap intervals.

3.4 Release engineering

Every candidate from RC1 to TF3 passed three checks before evaluation:

1. an independent standard-library parser reads all 6,980,503 rows and checks keys, value domains, horizon placement and that only the intended columns changed relative to the parent file;
2. a clean regeneration from the raw release and the saved weights, in a fresh pinned environment that rebuilds every cache, reproduces every queue label exactly and every continuous value to within 10^{-8};
3. the archive checksum (SHA256) is recorded, and each file is evaluated exactly once.

The later ES files change one column of a frozen, checksum-verified parent. Each was verified by recomputing its checksum and by a row-level comparison with that parent; ES2, for example, changes exactly 1,715,267 speed values and nothing else.

4 System architecture

Figure 3 shows the deployed pipeline. Three independent lanes produce the three row blocks of the submission file; Task 3 has no lane of its own and is computed by the evaluator from the state rows. The lanes share nothing at inference time. In particular, the offline state reconstruction never feeds a queue forecast, which keeps the Task 2 information set to its twelve history slots. Every learned artefact (trees, profiles, factor and covariance models, the queue classifier, the onset event pool and the trajectory library) is fitted on the training split, using the unmasked training layer as labels. The S10 and FS1 flow factors use only the visible cells of the split being reconstructed.

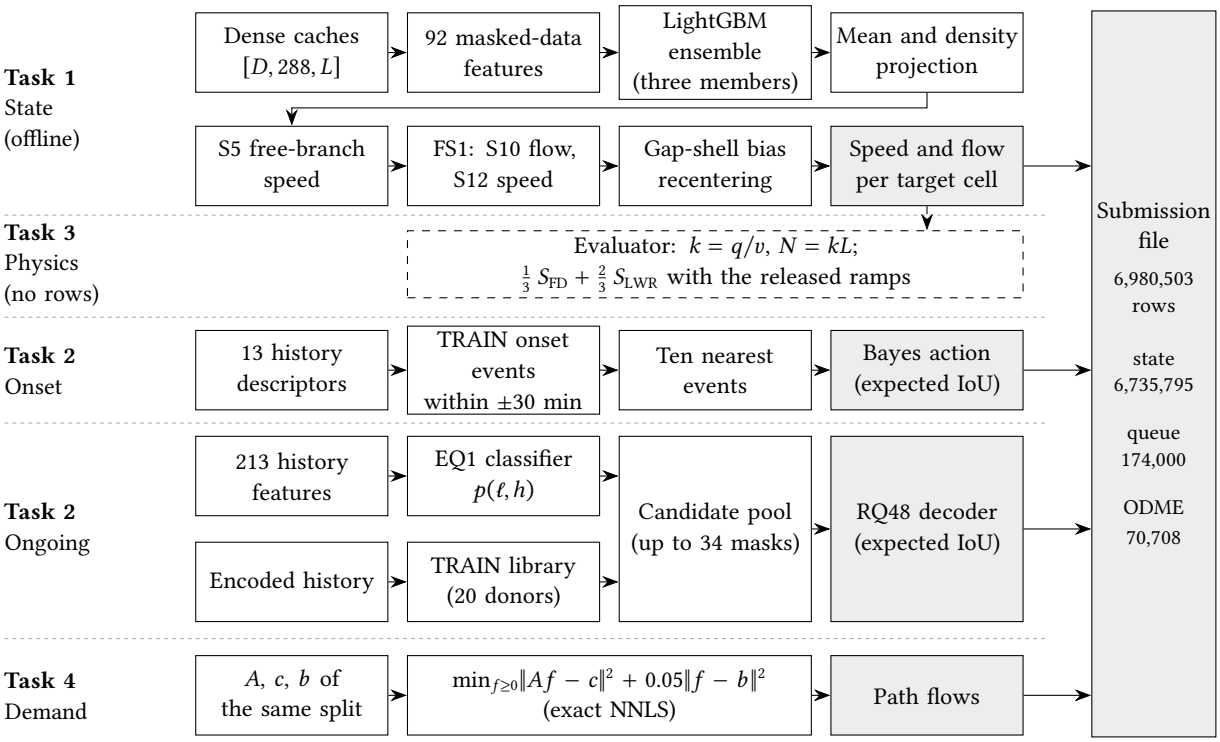


Figure 3. Deployed architecture of ES4. TF3 is identical except that it has no gap-shell recentering stage and uses the earlier Q46 classifier in place of EQ1. Solid arrows denote data flow at inference; the dashed arrow denotes the evaluator's own computation. No information flows between lanes.

4.1 Component lineage

The pipeline grew through single, measured changes. Table 5 lists each deployed component, the submission that introduced it and the change in validation score that it produced. Section B expands the identifiers.

Table 5. Components of the final submissions. Δ is the change in validation score of the submission that introduced the component. Two rows changed two components at once, and the RQ2 value follows from additivity (CS1 = J1 + RS1 state + RQ2 ongoing, Section 9).

Component	Task	Introduced in	Δ	TF3	ES4
Exact split-specific NNLS demand	ODME	T4-v1	+0.03158	✓	✓
Residual GBDT on temporal interpolation	State	V1	+0.12691	(RS1)	(RS1)
Joint-mask onset analogue	Onset	J1	+0.01861	✓	✓
Forward-filled persistence	Ongoing	U2	+0.02021	candidate	candidate
RS1 residual model with density reconciliation	State	RC2	+0.00974	✓	✓
Trajectory-analogue decoder (RQ2)	Ongoing	CS1	+0.00414	✓	✓
S5 speed correction and Q19 features	State, ongoing	CT1	+0.00039	✓	✓
S10 factor flow and S12 covariance speed	State	CT2	+0.00151	✓	✓
FS1 gate and Q46 first-crossing classifier	State, ongoing	CF1	+0.00449	✓	FS1 only
RQ48 quarter-prior decoder	Ongoing	CF2	+0.00469	✓	✓
Three-member sample ensemble (TS3)	State	TF3	+0.00034	✓	✓
All-month ongoing classifier (EQ1)	Ongoing	ES1	+0.00039		✓
Gap-shell recentering, shells 1, 2 and 3–6	State	ES2 to ES4	+0.00157		✓

5 Task 1: state reconstruction

5.1 Formulation

For one split and panel let $v_{d,t,\ell}$ and $q_{d,t,\ell}$ be speed and flow on day d, five-minute slot $t \in \{0, \dots, 287\}$ and link ℓ (links in topological order). We store every channel as a dense $[D, 288, L]$ grid. A cell is an *anchor* when it is eligible, finite and not a Task 1 target; target cells carry no speed, flow or occupancy. For a target cell let (t^-, t^+) be the nearest anchors on the same link and day, with gaps $g^{\pm} = |t - t^{\pm}|$, and let

$$\tilde{v}_{d,t,\ell} = \frac{g^+ v_{d,t^-,\ell} + g^- v_{d,t^+,\ell}}{g^- + g^+} \tag{5}$$

be same-day linear interpolation (edge values are held). Rather than predict v and q directly we predict the *residuals* $v - \tilde{v}$ and $q/n_\ell - \tilde{q}/n_\ell$, where n_ℓ is the lane count, so that flow is learned in the per-lane units of Equation (2). Interpolation is a strong prior on short gaps; the residual model only has to learn when and how it is wrong.

5.2 Baselines

The organizer baseline predicts a weekday × time-of-day profile per link and scores $S_{\text{state}} = 0.6929$ on the validation month. Same-day interpolation is far stronger, because most gaps are short: of the 3.44 million February training targets, 3.14 million have both anchors within six slots (Figure 5). The adaptive smoothing method of Treiber and Helbing [11] (ASM) supplies a third estimate. It averages anchors with two exponential kernels aligned with the free-flow and congested characteristics,

$$\phi_c(\Delta x, \Delta t) = \exp\Big(- \frac{|\Delta x - c\,\Delta t|}{\sigma_x} - \frac{|\Delta t|}{\tau} \Big), \qquad c_{\text{free}} = 100 \text{ km/h},\ c_{\text{cong}} = -18 \text{ km/h}, \tag{6}$$

with $\sigma_x = 0.8$ km and τ one slot, over ± 3 slots and ± 6 links, and blends the two estimates with weight $w = \frac{1}{2}\big(1 + \tanh((60 - \min(v_{\text{free}}, v_{\text{cong}}))/20)\big)$ on the congested one. The cell itself is excluded, so the estimate at an anchor is leave-one-out.

5.3 Features

Every feature is computed from the masked layer, static network files and TRAIN-fitted parameters (Table 6). The groups encode four kinds of evidence: the temporal anchors on the same link, the same-time state of neighbouring links and their own interpolation errors ("innovations"), multiscale and low-rank summaries of the visible field, and static context. The innovation of a neighbouring anchor, $\lambda_{t,\ell'} = v_{t,\ell'} - \tilde{v}^{\text{loo}}_{t,\ell'}$, measures how far that link departs from its own interpolant at the same time, which is direct evidence of a passing wave.

5.4 Learner

Three LightGBM [12] regressors are trained per member on pooled panels with the panel as a categorical feature: the speed residual and the per-lane flow residual under squared error, and density k under absolute error. Settings are fixed and were never tuned on evaluation months (Table 7). The final fit samples up to 600,000 eligible target cells per panel, six million in total, from the 31,783,763 eligible TRAIN target cells.

Density reconciliation. The three predictions $(\hat{v}, \hat{q}, \hat{k})$ need not satisfy $q = kv$. We take a half step of the first-order, error-weighted projection onto that constraint. With $a = -\hat{q}/\hat{v}^2$, $b = 1/\hat{v}$, $r = \text{clip}(\hat{k} - \hat{q}/\hat{v}, -10, 10)$, $\sigma_v = 1.5$ km/h, $\sigma_q = 30$ veh/h/lane and $D = a^2\sigma_v^2 + b^2\sigma_q^2$,

$$v \leftarrow \text{clip}\Big(\hat{v} + \tfrac{1}{2}\,\frac{r\,a\,\sigma_v^2}{D},\ 5,\ 140\Big), \qquad q \leftarrow \max\Big(0,\ \hat{q} + \tfrac{1}{2}\,\frac{r\,b\,\sigma_q^2}{D}\Big). \tag{7}$$

Table 6. The 92 inputs of each state model. "Neighbour" means a link $\ell \pm k$ at the same slot. Factor features are reconstructions from rank-8 and rank-24 temporal factor models fitted on TRAIN.

Group	Features	Count
Temporal anchors	interpolated v, q; previous and next anchor values and gaps; previous occupancy	9
Adaptive smoothing	ASM speed and flow, free-flow and congested kernel estimates	4
Profiles and ramps	weekday × slot TRAIN profile of v and q (own day excluded); on- and off-ramp flow	4
Neighbour state	v, q and their innovations at $\ell \pm 1, \pm 2, \pm 3$; innovations of $\ell \pm 1$ one slot before and after	28
Static and calendar	lanes, free-flow speed, capacity, length, position; slot, weekday, regime, observation rate; panel	11
Multiscale speed	interpolation, gap, rank-8 and rank-24 factor fit, four box smooths (1 or 3 slots × 7 or 21 links), near links $\pm 1, \pm 2$	12
Multiscale flow	the same twelve for per-lane flow	12
Multiscale occupancy	interpolation, four box smooths, near links $\pm 1, \pm 2$	9
Density and quality	interpolated q/v, $\tilde{q}/\tilde{v}$; `pct_observed` of the target cell	3
Total		92

Table 7. State-model settings, identical for every member. There is no early stopping.

Setting	Value	Setting	Value
Boosting rounds	800	Feature fraction	0.9
Leaves	255	Bagging fraction (every round)	0.8
Learning rate	0.05	L_2 penalty	2
Minimum leaf size	150	Rows per panel and member	$\leq$ 600,000
Objectives	squared error for v and q residuals; absolute error for density		

The scales distribute the correction in proportion to each channel's typical error, so a density disagreement moves flow more than speed.

5.5 Free-branch corrections

On the free-flow branch the residual model's errors are small but spatially correlated. Three statistical corrections act only there; none enforces the diagram or conservation exactly.

- **S5, speed.** A rank-8 spatial covariance of free-flow speeds, fitted on June–November TRAIN, conditions each masked link on the same-time observed links. It applies where $\hat{v} > 0.92\, v_f$, $\hat{q} < 0.8$ capacity and the link had at least 100 free-flow observations.
- **S10, flow.** A rank-12 factor model of transformed per-lane flow and occupancy is fitted to one evaluation month's *visible* free-flow cells, with link-wise centering, channel scaling, and diagonal autoregressive penalties at lags 1–3 on the temporal factors that never cross a day boundary (24 alternating ridge updates). The temporal penalty follows temporal regularized matrix factorization [13]. This is legal offline adaptation: target channels are absent, and no label is read.
- **S12, speed.** A full spatial covariance of June–November TRAIN free speeds, projected to positive definite form with 5 % diagonal shrinkage, gives the conditional mean of the missing links M given the observed links O at the same time,

$$v_M^{\text{cond}} = \mu_M - P_{MM}^{-1} P_{MO} \left(v_O - \mu_O \right), \qquad P = \Sigma^{-1}. \tag{8}$$

The deployed gate FS1 widens the region to $v_0 > 0.85\, v_f$ and $q_0 < 0.9$ capacity, evaluated on the S5

output (v_0, q_0), and blends at half strength: $q = \frac{1}{2}q_0 + \frac{1}{2}q^{\text{factor}}$ and $v = \text{clip}(\frac{1}{2}v_0 + \frac{1}{2}v^{\text{cond}}, 5, 140)$. On February the fixed half-strength arm beat its predecessor on every panel (joint proxy gain +0.000156). On the validation month S10/S12 added 0.00151 (CT2 against CT1, state only); FS1 shipped together with a queue change.

5.6 A fixed three-member sample ensemble

A single tree ensemble fitted on one random sample of six million cells gives a slightly different field than a fit on another sample. For member predictions p_m, their mean $\bar{p}$ and any target y,

$$\frac{1}{M}\sum_{m=1}^{M}(p_m - y)^2 - (\bar{p} - y)^2 = \frac{1}{M}\sum_{m=1}^{M}(p_m - \bar{p})^2 \;\geq\; 0, \tag{9}$$

so averaging improves on the mean member by exactly the members' spread [14]. TS3 averages three members: the frozen RS1 regressors and two refits with seeds 20261104 and 20261204, each drawing up to 600,000 eligible rows per panel without replacement. The raw speed and flow predictions are averaged with equal weight and Equation (7) and the free-branch corrections are applied once, to the average. No member was dropped and no weight tuned after a score was seen.

The protocol of Section 3 was registered before fitting. Table 8 shows the development and confirmation results against a control that reproduced the parent exactly; the rule was then refit on all nine months and shipped as TF3, which raised the validation score by 0.00034.

Table 8. TS3 against its exact parent on held-out TRAIN months (models fitted on June–November). All ten panels improved in every month. Training proxies, not validation scores.

Month (role)	ΔS_{state}	$\Delta S^{\text{proxy}}_{\text{phys}}$	$\Delta(0.35\,S_{\text{state}} + 0.15\,S^{\text{proxy}}_{\text{phys}})$	Panels improved
December (development)	+0.00032340	+0.00172636	+0.00037214	10 / 10
January (development)	+0.00033570	+0.00178485	+0.00038522	10 / 10
February (confirmation)	+0.00025537	+0.00152578	+0.00031825	10 / 10

5.7 Short-gap link-bias recentering

Observation. On February training targets, the same-day interpolant Equation (5) is almost unbiased *per link* when both anchors are close: the mean absolute per-link bias of $\tilde{v} - v$ is about 0.02 km/h on one-slot gaps. The deployed reconstruction is not. On March, ES1's per-link means sit 0.77–1.54 km/h (RMS across links, by panel) from the same interpolant on one-slot cells. A model fitted on June–February carries a month-level offset into a new month; the interpolant, built from that month's own anchors, does not.

Estimator. The *gap shell* $G_{[a,b]}$ is the set of target cells that have both anchors and satisfy $a \leq \max(g^-, g^+) \leq b$. Because the interpolant's link bias on a passing shell is negligible,

$$\mathbb{E}\big[\hat{v} - v \,\big|\, \ell, G\big] \;\approx\; \mathbb{E}\big[\hat{v} - \tilde{v} \,\big|\, \ell, G\big] \;=\; \delta_\ell, \tag{10}$$

and the right-hand side needs no label. For each split, panel and link with at least 100 shell cells we compute δ_ℓ as the mean of $\hat{v} - \tilde{v}$ over the shell and set $v \leftarrow \text{clip}(\hat{v} - \delta_\ell, 5, 140)$. This removes the link-mean component of the error, MSE = bias^2 + variance, and keeps the model's within-link structure; it never replaces the model by the interpolant, which loses on hard cells (Section 5.8).

Gates. A shell passes on February when the interpolant's mean absolute link bias is below 0.15 km/h and a bias correction estimated on earlier months does not raise February RMSE by more than 0.02

km/h. A passing shell is written to a panel and split only if the RMS link shift is at least 0.40 km/h and no link moves by more than 15 km/h. Flow was not recentered: its February link bias (3–10 veh/h) is negligible against an RMSE near 200 veh/h, and the March link offset of the flow interpolant (10.9 veh/h RMS) missed its 20 veh/h bar.

Table 9. Gap-shell results. February columns are training proxies; "speeds changed" counts March and April cells; Δ is the change in validation score of the submission that wrote the shell.

Shell (slots)	Feb. link bias	Feb. RMSE, raw / corrected	Speeds changed	Mean $\lvert\Delta v\rvert$	Δ
1	0.020	unchanged	1,715,267	0.962	+0.00039 (ES2)
2	0.028	2.560 / 2.560	1,918,477	0.960	+0.00044 (ES3)
3	0.039	2.842 / 2.842	1,186,051		
4	0.059	3.085 / 3.086	671,422	0.954	+0.00074 (ES4)
5	0.082	3.423 / 3.425	319,161		
6	0.108	3.562 / 3.565	15,289		
7	0.143	3.646 / 3.651	0		not written
8	0.184		0		failed
9–12	0.196		0		failed
13–24	0.309		0		failed
25–48	1.451		0		failed

Across the three submissions the edit changed 5,825,667 speed values by about 0.96 km/h on average (RMS 1.25 km/h, maximum 6.24 km/h) and raised the validation score by 0.00157 in total. It is a small gain from a simple, label-free statistic, and it is the only evaluation-month adaptation of speed in the final submissions.

5.8 Results and error analysis

Table 10 and Figure 5 compare the RS1 residual model with interpolation on February, held out from a fit on June to November. Speed RMSE falls from 2.75–2.98 to 1.46–1.66 km/h and per-lane flow RMSE from 44.5–47.6 to 30.4–31.1 veh/h, giving a state piece of 0.940–0.945 against 0.899–0.907. The 50 % mask costs the model only 0.005 in the piece relative to the 20 % mask. The advantage holds at every gap length: the model roughly halves interpolation error from one-slot gaps (1.41 against 2.37 km/h) to gaps of 13 to 30 slots (1.99 against 3.98 km/h).

Table 10. February state error by masking regime (held out from a fit on June to November; training proxy).

		Speed RMSE (km/h)		Flow RMSE (veh/h/lane)		State piece	
Regime	Cells	Interp.	Model	Interp.	Model	Interp.	Model
R1 (20 %)	555,352	2.748	1.460	44.5	30.4	0.9066	0.9451
R2 (30 %)	902,718	2.839	1.553	46.2	30.9	0.9032	0.9428
R3 (50 %)	1,985,874	2.975	1.661	47.6	31.1	0.8993	0.9403

Where the error is. Error concentrates where a wave passes inside the gap. Among February cells with both anchors within six slots, 138,195 have anchors that disagree by at least 20 km/h or an interpolant at most $0.6\,v_f$. There interpolation reaches 10.57 km/h RMSE and the model 4.62 km/h, better on all ten corridors. Same-time neighbours do not substitute for temporal anchors: on 735,668 cells where both adjacent links are observed, their average scores 6.37 km/h against the model's 1.44, and switching to the neighbours only when the temporal anchors disagree by 20 km/h still scores 1.60 km/h. The model already combines both sources better than any fixed rule we tried.

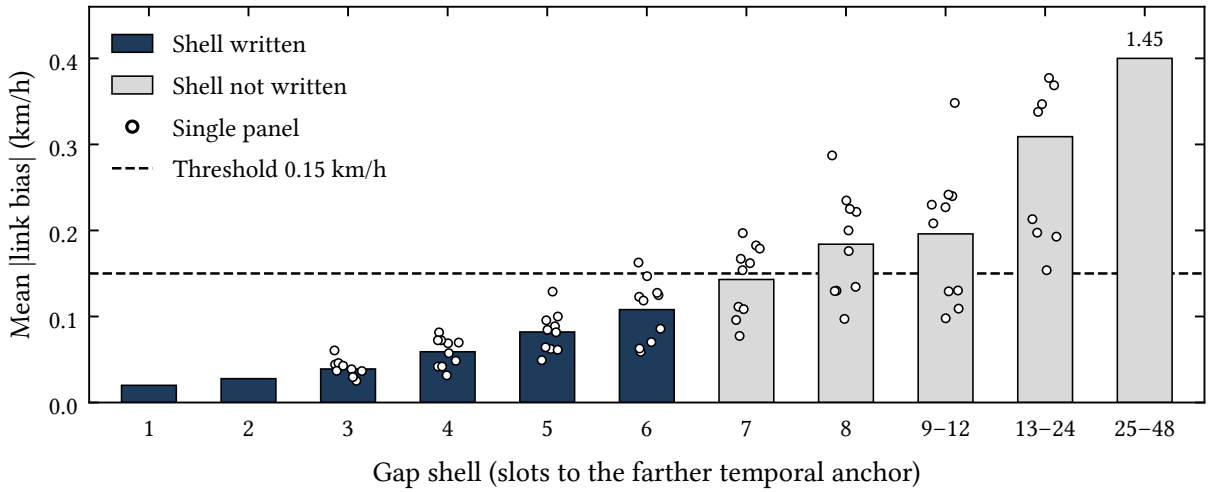


Figure 4. February mean absolute link bias of the same-day interpolant by gap shell. Bars pool all panels; circles show individual panels; the dashed line is the 0.15 km/h acceptance threshold. Shells 1 to 6 were written (in ES2, ES3 and ES4). Shell 7 passed the test, but no March or April link reached the 100 cells required by the writing rule. The bar for shells 25 to 48 is truncated.

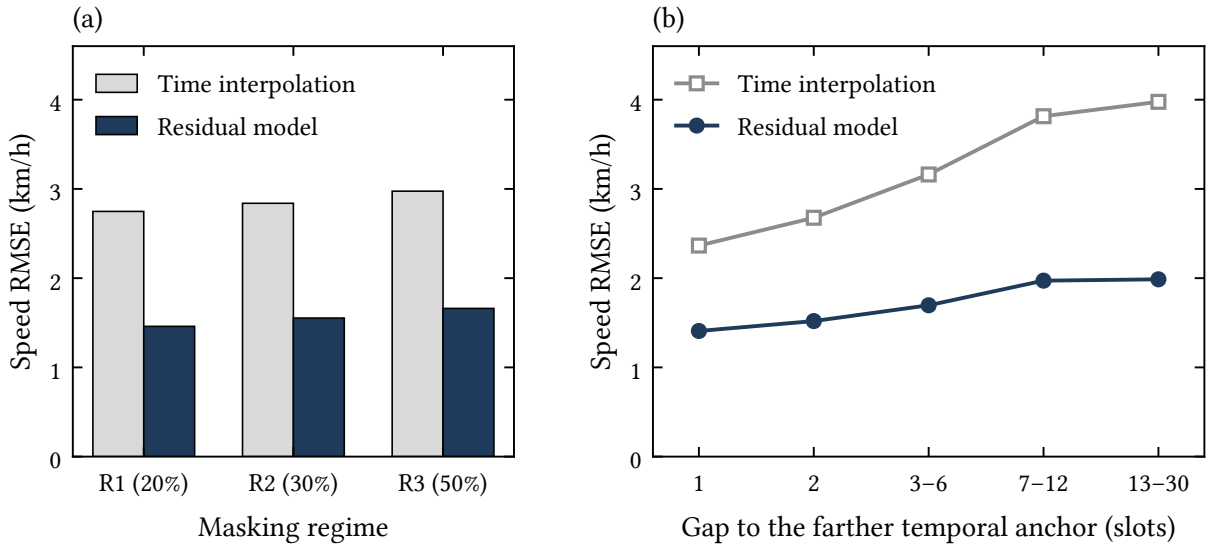


Figure 5. February speed error of the residual model and of same-day interpolation (a) by masking regime and (b) by the distance to the farther temporal anchor, in five-minute slots. Cell counts by gap class are 804,794, 963,903, 1,371,721, 214,200 and 34,936.

Validation evidence. Replacing the first residual model by plain interpolation (L0 against U2) lowered the validation score by 0.00867, and the RS1 model added 0.00974 over the first model (RC2 against RC1). Month-blocked cross-validation on training months had predicted about 0.021 for the first model over interpolation in the same state-and-physics block, $0.35\,S_{\text{state}} + 0.15\,S_{\text{physics}}$; the validation month showed 0.0087. The model transfers to a new month, but less than half of its cross-validated advantage survived. For this reason every later state change was tested on held-out months, and we treat state proxies as optimistic by a factor of about two.

6 Task 3: physics consistency

Task 3 has no rows. Its 15 % weight is earned or lost by the Task 1 answer, through the two terms of Equation (4). This section explains why we ended up *not* imposing physics on the reconstruction, after testing it many ways.

6.1 The diagram term is saturated

S_{FD} compares the submitted per-lane flow with the triangular diagram at the submitted density. Any field whose speed and flow are jointly plausible satisfies it. Scored against the released `fd_parameters.csv`, ES4's own speed and flow reach 0.984–0.994 on all twenty panel-splits of validation and private, with a mean absolute flow residual of 5–15 veh/h and essentially no clipping. Exact observations score about 0.986 on the training months. A projection of the field onto the diagram buys about +0.002 in S_{physics} and costs speed accuracy, so it was not used.

6.2 The conservation term cannot be measured locally

S_{LWR} checks $N_{t+\Delta t} - N_t$ against boundary flows that the organizers compute from their own state. Those flows are withheld. Our local scorer is a line-by-line equivalent of the official one: given the same boundary-flux file, the two agree to four decimal places on eight test fields and four panels. With the flux replaced by the observed accumulation change, it sits near the exact-answer ceiling (the organizers' perfect answer scores 0.9636 in S_{physics}, about 0.95 in S_{LWR}), so it is optimistic and compresses differences between methods. We therefore used it only as a non-regression check next to S_{state}, never as the objective.

6.3 Imposing physics lost every time

We tested the most natural ways to make the reconstruction obey the kinematic-wave model [9, 10, 15]. Each was frozen on fitting months and scored once on held-out months (Table 11). The first four rows use one corridor (D12_I5_N, all 2,583,593 October to February targets, with same-day interpolation as the control); the Newell rows use February on all ten panels, with the deployed residual model as the control.

6.4 Why physics loses here

The results share one explanation. The masked holes are short in space and time. On D12_I5_N, of 1,159,377 ramp-free target cells with a detector on both sides, 494,175 sit in one-link holes, 345,813 in two-link holes and 295,706 in holes of three to five links; no hole is longer than twenty links. The information about a missing cell is mostly in its own detector a few minutes earlier and later. A fine-step solver moves a wave across the whole hole within one five-minute bin, so its five-minute aggregate becomes a *spatial* reconstruction, and spatial reconstruction is the weaker estimator (corridor interpolation 0.831 against temporal 0.904; the solver scores 0.813). Beyond that, incidents and capacity drops in the generator are unobserved forcing; a detector's five-minute flow is not the cell-boundary flux of the conservation check; and the conservation signal itself is small (about 0.75 % of the vehicles on a link) relative to the measurement noise.

In summary, the physics weight of this benchmark is best served by an accurate state rather than by a physical model of the state. The diagram term is satisfied by any accurate field, and the conservation term moved together with state accuracy in every held-out test; TS3, for example, raised the local physics proxy by 0.0015 to 0.0018 while raising S_{state}. Imposing the kinematic-wave model never outperformed temporal evidence on held-out months.

Table 11. Physics-based reconstruction experiments. "Piece" is the state score of Equation (2) for the method and for its control. All values are held-out training proxies; none of these variants is used in a final submission.

Method	Piece	Control	Outcome
Godunov LWR, 0.245 s step, 1,224 substeps per bin, released diagram, visible cells pinned [5, 6]	0.8127	0.9045	congested speed RMSE 20.68 vs 8.10 km/h; gate +0.005 failed
Corridor (spatial) interpolation of speed and flow	0.8307	0.9045	speed RMSE 6.59 vs 2.93 km/h
Diagram inversion of spatially interpolated density	0.7449	0.9045	flow RMSE 104.3 vs 42.1 veh/h/lane
Steady flux $\min(\text{sending}, \text{receiving})$ on 1,159,377 ramp-free holes	0.8510	0.9074	congested speed RMSE 18.60 vs 7.98 km/h
Time-interpolated speed with corridor-interpolated flow	+0.0037 (mean gain)		lost on two corridors; local physics lower on seven
Newell lower envelope of cumulative counts [7]	0.4248	0.9456	speed RMSE 79.03 vs 1.60 km/h
Newell envelope only on congested cells	0.8292	0.9456	speed RMSE 7.36 km/h
Flow nudge of at most 40 veh/h/lane toward the envelope	0.9408	0.9456	physics proxy 0.700 vs 0.743; projected −0.0081 total
Congested-branch diagram flow from the interpolated speed			flow RMSE 101.4 vs 63.3 veh/h/lane on 131,696 cells, worse on all ten panels
Occupancy density $k = 285 \cdot \text{occ}$ inverted through the diagram			below interpolation on all ten panels
Flow and ramp balance regression on consecutive visible pairs			held-out correlation 0.0009–0.0094 (threshold 0.5)

7 Task 2: queue forecasting

7.1 Formulation

A window w has origin T and history H_w: speed, flow, occupancy, eligibility and ramp counts on the twelve slots $T-60, \dots, T-5$ (Figure 2). We must output a binary field $a \in \{0,1\}^{6\times L}$ over links and the six horizons. The decision that maximizes the expected score is

$$a^\star(H_w) = \arg\max_a \ \mathbb{E}\big[\mathrm{IoU}(a, Y) \,\big|\, H_w\big], \tag{11}$$

which has no closed form because IoU is a ratio over the whole field [2]. We approximate the expectation in two ways, writing J for an estimate of the Jaccard index (IoU). An *empirical* estimate $J_{\text{emp}}(a)$ averages the IoU of a against the complete outcomes y_j of analogous training windows [3]. A *model* estimate uses per-cell probabilities p_i and the ratio of expected intersection to expected union,

$$J_p(a) = \frac{\sum_i a_i p_i}{\sum_i a_i + \sum_i p_i - \sum_i a_i p_i}. \tag{12}$$

This ratio of expectations is not the exact expected IoU, but it rewards the right trade-off between size and confidence, which a per-cell threshold does not.

7.2 What the window selection implies

Two properties of the selector (Section 2.5) decide the shape of a good forecaster.

- **Onset queues appear at the last step.** In the 40 published training onset windows the median first crossing of $0.6\,v_f$ is at horizon 6. The head of the queue is usually a recurrent bottleneck whose location a training prior already contains: the top three links of a frequency prior include it in 79 % of events. The size is the hard part. With a perfect head and an unknown extent a contiguous block scores about 0.62 latent IoU; with the true extent, 0.905. Extent varies from one to eight links and we found nothing in the legal hour that predicts it (correlation 0.05 with daily demand, 0.10 with hourly demand).
- **Ongoing windows are those persistence gets wrong.** Windows with persistence IoU above 0.9 are excluded. In 31 of the 40 published ongoing windows the queue began before $T-60$, so most queues are mature, and their evolution over the next 30 minutes (clear, hold or grow) is the forecast.

7.3 Ongoing queues: start by not dropping sensors

The organizer's persistence baseline repeats the queue state of slot $T-5$ using eligible cells only. Between 23 and 24 % of links have a missing or ineligible $T-5$ reading in every split, and persistence then predicts "not queued" there. Forward-filling the last *released* value of each link over the twelve slots, ineligible readings included, keeps those links. On the validation month this raised ongoing IoU from about 0.603 to 0.738 (U2 against U1, an ongoing-only change of 0.02021 in the composite score). A diagnostic that set ongoing queues from the state at slot T (P-slotT) reached only about 0.729. On the training windows slot T adds 0.003 ± 0.002 latent IoU over the forward-fill. What mattered was not the extra slot but retaining the sensors that were already reporting.

7.4 Onset: a joint-mask analogue

Per-link classifiers decide each link independently and so cannot represent the joint size and shape of an onset. The J1 rule instead chooses a complete mask from the outcomes of similar training events.

1. **Event pool.** Training origins that meet the legal onset definition (no queue in $T-60 \dots T+25$, a queue at $T+30$, history coverage at least 0.7), within ±30 minutes of the query origin. Candidates are tiered: same day type (weekday or weekend) and local time first, then any day type, then any slot.
2. **Similarity.** Thirteen descriptors of the legal hour: total flow over capacity (last, mean, change), the mean speed-to-threshold ratio (last, mean, change), the minimum ratio, the eligibility rate, total ramp flow (last, mean, change), and the sine and cosine of the time of day. Distance is the standardized RMS difference, and the ten nearest events are kept.
3. **Action set.** Each neighbour's observed $T+30$ mask, plus the top-k links of the neighbours' marginal frequency for $k = 1, \dots, 15$.
4. **Bayes action.** The action with the highest mean IoU against the ten neighbour outcomes on their known cells, placed at $T+30$ only; horizons 1–5 are empty.

On the validation month the twelve-slot LightGBM onset classifier scored about 0.626 IoU, a frequency prior 0.450 and J1 0.750 (Table 14). J1 has supplied the onset forecasts of every later candidate, including both final submissions. Its 0.75 is close to what the training analysis suggests twelve slots allow (0.62 with unknown extent, 0.905 with the true extent).

7.5 Ongoing queues: classifier, analogue library and decoder

Training population. A training origin is *difficult* when its observed six-step future has queued cells, its denoised origin state has a queue, and six-step persistence scores at most 0.9 IoU on it, mirroring the selector. A *first crossing* is a difficult origin whose predecessor was not difficult. Q46 was fitted on 2,093 first crossings from July–November (at most 300 per panel). EQ1, in ES4, keeps every one of them and adds up to 240 date-diverse first crossings per panel from June, December, January and February: 3,766 windows and 3,172,453 rows. Future training observations are labels only, never inputs.

Features. 213 inputs per link and horizon, all from the twelve legal slots (Table 12). Beyond lags and summaries, two groups carry the traffic physics. Distance bands summarize the speed ratio, flow ratio and density 1, 2, 4, 8 and 16 km upstream and downstream. Characteristic features read the state along the free-flow and backward-wave characteristics through each link, at ages of 0, 15 and 30 minutes, which is where a wave that will reach the link within the horizon is visible now.

Table 12. The 213 inputs of the ongoing-queue classifier.

Group	Features	Count
Lags	speed and flow at lags 1, 2, 3, 4, 6, 9, 12 slots	14
Summaries	recent, previous, early, min, max, std, slope, trend, coverage, age; for speed and flow	20
Neighbours	last value and trend at links $\pm1, \pm2, \pm4$; for speed and flow	24
Corridor	corridor mean, minimum and trend; for speed and flow	6
Queue context	queued neighbours at $\pm1, \pm2, \pm3$, queued count and its change, nearest queue up- and downstream	10
Density	q/v density, last value, trend, acceleration, distance to the queue cut	5
Distance bands	speed ratio, flow ratio and density at $\pm1, 2, 4, 8, 16$ km: recent, trend, extreme	90
Characteristics	free-flow and wave characteristics, both directions, ages 0/3/6 slots; speed, flow, density	36
Static and calendar	free-flow speed, per-lane capacity, lanes, length, per-lane jam density; slot, weekend, horizon	8
Total		213

Classifier. A LightGBM binary model with 63 leaves and learning rate 0.05, equal weight per panel and per window, and the horizon as an input, so one model covers all six steps. Q46 has 400 trees; EQ1 has 800.

Analogue library. Every training origin, at a stride of two slots, is encoded by its twelve-slot speed and flow ratios and its complete six-step future. For a query, donors are restricted to ±2 hours of time of day and the same weekend flag when at least 20 such donors exist. The distance is the mean squared difference of the speed-ratio history, with links currently below $0.85\,v_f$ weighted thirteen times, plus 0.4 times the flow-ratio distance, plus 0.15 times one minus the Jaccard overlap of the current queue sets. The 20 nearest donors are kept.

Candidate pool. The decoder chooses among at most 34 complete masks: the analogue's own choice; persistence; the expected-count decoding of p; $p \geq 0.4$ and $p \geq 0.5$; the 20 donor futures; the donors' marginal frequency thresholded at 0.25, 0.40, 0.55 and 0.70; the union and intersection of the analogue choice with the decoded p; and the expected-count decoding of α marginal + $(1 - \alpha)\,p$ for $\alpha \in \{0.25, 0.5, 0.75\}$.

Decoder (RQ48). With $J_{\text{emp}}(a)$ the mean IoU of a against the donor futures on their known cells, the deployed choice is

$$\hat{a} = \arg\max_{a \in \mathcal{A}} \Big[0.25\, J_{\text{emp}}(a) + 0.75\, J_p(a) \Big]. \tag{13}$$

The earlier rule used equal weights. Four weightings were registered before development; the quarter weight had the largest minimum gain across the March and April development samples, passed its single confirmation (+0.0335 and +0.0300 on even dates, date-clustered 95 % intervals [0.013, 0.054] and [0.012, 0.051]), and raised the validation score by 0.00469, which corresponds to 0.031 ongoing IoU.

7.6 Results and the selection gap

Table 14 gives the validation evidence. Ongoing IoU rose from about 0.603 (organizer persistence) to 0.738 (forward-fill) and 0.766 (first analogue decoder). Two later steps also changed the state and cannot be separated; after them RQ48 added 0.031 and EQ1 0.003. If the two state corrections in those mixed steps contributed between nothing and their full held-out proxy value, ES4's ongoing IoU on the validation month lies between 0.827 and 0.832. Onset IoU remained at 0.750.

Figure 6 shows where the remaining queue error lies. On the 40 published ongoing windows, the analogue's choice scores 0.836 while the best mask in its own candidate set scores 0.934. On 146 visible March windows the deployed decoder scores 0.813 against a best-in-set 0.875, and the analogue's choice alone ties persistence (0.718). On 160 hard November windows the best whole-window action scores 0.946 and the best per-link combination 0.967, against 0.807 for the deployed whole-window rule. The candidate set is therefore adequate; the selection within it is not. None of the frozen selection rules we tried reached its pre-set gain of 0.02 to 0.03 IoU on held-out windows: a ridge reranker (+0.007), an expected-overlap-minus-disagreement ranker (+0.009), a cell majority of the 20 donors (+0.004), a neighbour-bit classifier (−0.010), per-link nearest-neighbour copies (−0.022), odd-date panel weights and a March-fit reranker (Section 10).

In summary, queue forecasting in this benchmark is a decision problem over complete masks. The two largest gains came from retaining information (forward-filling dropped sensors) and from respecting the joint structure of the outcome (choosing whole onset masks by expected IoU). Better per-cell probabilities moved the score less than the choice among whole masks, and that choice, rather than the candidate set, is where the remaining error lies.

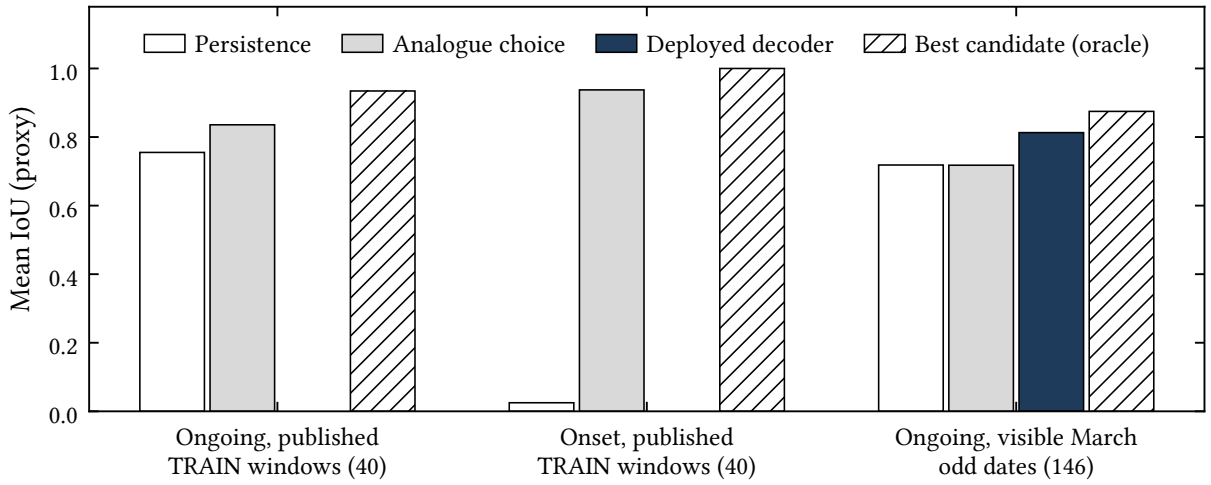


Figure 6. Mean IoU of persistence, the analogue's choice, the deployed decoder and the best mask already present in the candidate set (identified with the label after the choice; a diagnostic, not a forecast). Published training windows use latent labels and visible March windows use observations. The deployed decoder was not evaluated on the published windows.

8 Task 4: path-flow and OD estimation

8.1 Formulation

Each panel ships a path set (one path per upstream-to-downstream zone pair, Table 2), the path-link incidence $A \in \{0,1\}^{|\mathcal{L}|\times|\mathcal{P}|}$, the split's link counts c and a weak path-flow prior b. We solve, separately for validation and for private,

$$f^\star = \arg\min_{f\geq 0} \|Af - c\|_2^2 + \lambda \|f - b\|_2^2, \qquad \lambda = 0.05, \tag{14}$$

a generalized least-squares estimator in the tradition of Cascetta [16]. Writing it as one non-negative least-squares problem,

$$f^\star = \arg\min_{f\geq 0} \left\| \begin{bmatrix} A \\ \sqrt{\lambda}\, I \end{bmatrix} f - \begin{bmatrix} c \\ \sqrt{\lambda}\, b \end{bmatrix} \right\|_2^2, \tag{15}$$

lets the active-set algorithm of Lawson and Hanson [4] (SciPy's `nnls` [17]) return the exact optimum. There are 35,354 paths in total and at most 5,050 per panel, so the twenty solves are small. The result is byte-identical to the `public_reference.csv` that the organizer's own builder produces.

8.2 Comparison of estimators

We isolated Task 4 with submissions that differ from the organizer baselines only in their demand rows (Table 13). Because the other three tasks are identical, each composite score converts exactly to S_{ODME} through $S_{\text{ODME}} = 0.8359 + (S_{\text{total}} - 0.55343)/0.20$.

The organizer baseline solves the same problem against *training* counts and scores 0.836. Solving it exactly against the split's own counts and prior gives an implied 0.994. Every attempt to model the prior did worse. Analysis of the training split suggested a prior equal to the truth plus positive noise, but no parametric law we fitted, such as gravity or distance decay, explained it well enough to help. The entire Task 4 gain came from solving the correct system exactly.

With a weak prior and a regularizer chosen by the organizers, the scoring reference is itself a regularized solve. The best estimate is therefore the exact optimum of the same objective on the same

Table 13. Demand estimators evaluated with otherwise identical baseline rows. S_{ODME} is implied by the composite score.

Estimator	Implied S_{ODME}
Exact NNLS, $\lambda = 0.05$, on the split's own counts and prior	**0.994**
Entropy (KL) fit to the raw prior	0.843
Organizer baseline: the same solve against training counts	0.836
KL fit from a fitted gravity seed	0.622
Prior multiplied by a smooth distance correction	0.468

split's data; structure that is not in the objective (gravity, distance decay, entropy) only moved the answer away from the reference.

9 Analysis of validation results

9.1 Exact additive identification

The three row blocks of a submission are disjoint, and each task is scored only on its own rows. Writing $B = 0.35\,S_{\text{state}} + 0.15\,S_{\text{physics}}$ for the block that depends on the state rows,

$$S_{\text{total}} = B + 0.20\,S_{\text{ODME}} + 0.15\,\text{IoU}_{\text{onset}} + 0.15\,\text{IoU}_{\text{ongoing}} \tag{16}$$

holds exactly. Two submissions that differ in one block therefore measure the change of that block exactly, up to the five-decimal rounding of reported scores ($\pm 5 \times 10^{-6}$ per score, or about $\pm 7 \times 10^{-5}$ in an IoU). The organizers' baseline table (Table 3) anchors absolute values: persistence predicts nothing on onset windows, so its $S_{\text{queue}} = 0.3017$ is half of an ongoing IoU of 0.6034.

Chaining single-block contrasts from that anchor identifies the values in Table 14. The chain is consistent: rebuilding J1 from its identified parts gives 0.84461, exactly its reported score. Two steps (CT1 and CF1) changed two blocks at once, so the state and ongoing values after them are bounded rather than identified (Section 7.6). The split of B between S_{state} and S_{physics} is not identifiable.

Table 14. Component values identified from validation scores through Equation (16). Generated from the submission record by `docs/scripts/make_figures.py`.

Task	Quantity	Value	Identified from
Demand	S_{ODME}, organizer solve on TRAIN counts	0.8359	documented
	S_{ODME}, exact NNLS on the split's own data	0.9938	T4-v1 − BASE
State	B, weekday × slot historical mean	0.2957	documented
	B, time interpolation	0.4140	L0 − U2
	B, first residual GBDT (V1)	0.4226	P-persq
	B, RS1 residual GBDT	0.4324	RC2 − RC1
Onset	IoU, persistence (predicts nothing)	≈ 0	by construction
	IoU, first LightGBM, 13 slots incl. T (V1)	0.6393	V1 − P-noonset
	IoU, legal 12-slot classifier (U1)	0.6258	U1 − P-persq
	IoU, training frequency prior (U3)	0.4497	U3 − U2
	IoU, demand-history completion (CS3)	0.7116	CS3 − CS2
	IoU, joint-mask analogue (J1, kept to ES4)	0.7499	J1 − U2
Ongoing	IoU, organizer persistence	0.6034	documented
	IoU, first LightGBM, 13 slots incl. T (V1)	0.5922	P-noonset
	IoU, same model, unfiltered training windows	0.6144	V1-nh − V1
	IoU, state at slot T (diagnostic only)	0.7293	P-slotT − V1
	IoU, forward-fill of the last legal value (U2)	0.7381	U2 − U1
	IoU, RQ2 analogue decoder (CS1)	0.7657	CS1 identity
	IoU, origin analogue (CS2)	0.7397	CS2 − CS1
	ΔIoU, RQ48 quarter-prior decoder	+0.0313	CF2 − CF1
	ΔIoU, occupancy classifier RQ59	-0.0007	CF3 − CF2
	ΔIoU, all-month classifier EQ1	+0.0026	ES1 − TF3

Three observations follow. The residual state model added 0.127 to B over the historical-mean baseline, more than any other single change. Onset forecasting moved from zero to 0.750 IoU, worth 0.112 in the composite score. Ongoing forecasting, which received most of the research effort after the first week, moved from 0.603 to about 0.83, worth about 0.034.

9.2 Decomposition of the improvement

Table 15 decomposes the improvement from the organizer baselines to ES4 into sixteen measured steps. Each step is the difference between two validation scores; the RS1 and RQ2 steps follow from additivity.

Table 15. Measured steps from the organizer baselines to ES4.

Step	Change	Rows changed	Δ score	Cumulative
0	Organizer baselines (anchor)	all		0.55343
1	Exact split-specific NNLS demand	Demand	+0.03158	0.58501
2	Residual GBDT state reconstruction	State	+0.12691	0.71192
3	Legal onset classifier (U1)	Onset	+0.09387	0.80579
4	Forward-filled ongoing persistence (U2)	Ongoing	+0.02021	0.82600
5	Joint-mask onset analogue (J1)	Onset	+0.01861	0.84461
6	RS1 state model (RC2 − RC1)	State	+0.00974	0.85435
7	RQ2 ongoing analogue decoder (CS1, by additivity)	Ongoing	+0.00414	0.85849
8	S5 speed correction and Q19 features (CT1)	State, ongoing	+0.00039	0.85888
9	S10 factor flow and S12 covariance speed (CT2)	State	+0.00151	0.86039
10	FS1 gate and Q46 first-crossing classifier (CF1)	State, ongoing	+0.00449	0.86488
11	RQ48 quarter-prior decoder (CF2)	Ongoing	+0.00469	0.86957
12	Three-member sample ensemble (TF3)	State	+0.00034	0.86991
13	All-month ongoing classifier (ES1)	Ongoing	+0.00039	0.87030
14	Gap-1 speed recentering (ES2)	State	+0.00039	0.87069
15	Gap-2 speed recentering (ES3)	State	+0.00044	0.87113
16	Gap 3–6 speed recentering (ES4)	State	+0.00074	0.87187

9.3 Transfer of local proxies

Seventeen submissions changed the rows of exactly one task relative to an earlier file (Figure 7). Each was evaluated because a local proxy or a measured rationale favoured it. The contrast between the two panels of the figure is the most useful methodological result of this work.

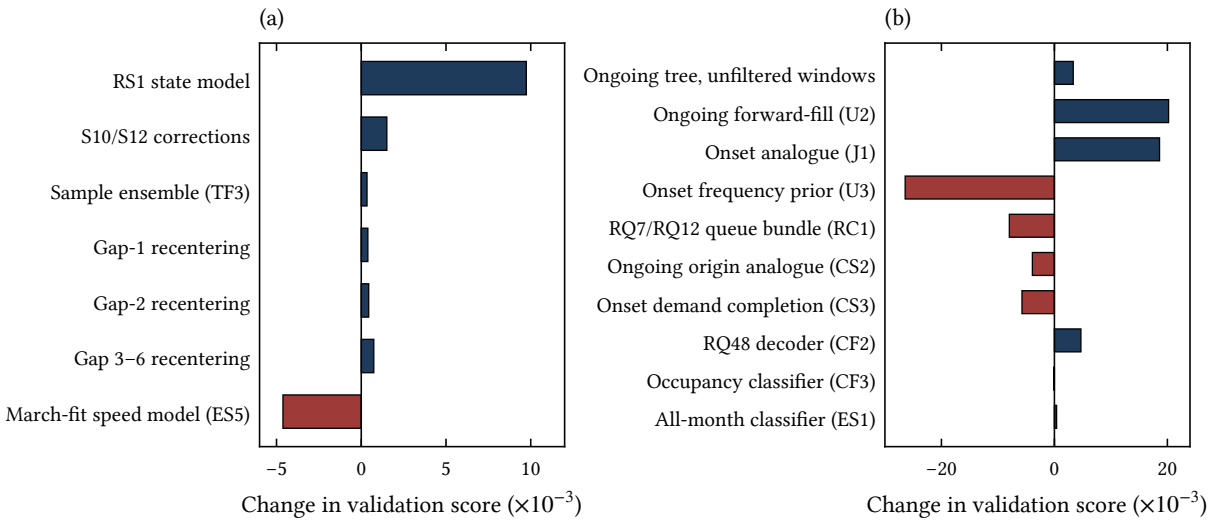


Figure 7. Change in validation score of every single-task change: (a) state-only changes, (b) queue-only changes.

- **State changes transferred.** All six state changes selected on held-out training months improved the validation score, though by less than the held-out proxies suggested (Section 5.8). The single state loss, ES5 (−0.00462), is the only model fitted and selected on visible cells of the evaluation month itself. It lowered visible-cell speed RMSE from 2.545 to 2.006 km/h on every corridor, yet the visible cells were not a stand-in for the masked holes.
- **Queue changes did not.** Five of ten queue-only changes lowered the score. CF3's occupancy classifier gained 0.0074 to 0.0085 ongoing IoU on visible March proxies and lost 0.0007 on validation; the onset frequency prior kept its skill leave-one-month-out on training windows and lost 0.176 onset IoU on validation.

Two facts explain the asymmetry. State scores average over millions of cells, so a real improvement is visible; queue scores average over 80 windows per split. Moreover, the visible-cell and published-window proxies differ in kind from the hidden targets: visible pseudo-events are noisy observations rather than latent labels, and the published windows are 80 events from ten days in June.

9.4 Sampling noise and the private month

The organizers estimate that queue-window sampling alone moves S_{total} by about 0.0126 (one standard deviation). The private month has seven incidents against five in the validation month, at different places and times. Differences of a few thousandths between files are therefore not a reliable forecast of the private ordering, and the state reconstruction is the most stable part of any composite score.

9.5 Choice of the final submissions

Two submissions are scored on the private month. They were chosen for complementary reasons.

- **TF3** is the most defensible artefact: every component was selected on held-out training months under pre-registered tests, and the whole file regenerates from the raw release with one command (Section 12).
- **ES4** adds the two later changes that improved the validation score, the all-month ongoing classifier and the label-free short-gap recentering, and shares TF3's onset, flow and demand rows exactly.

ES5, the only later file, was not selected: it lowered the validation score, and its model was fitted on the evaluation month.

10 Negative results

Most of what we built was not used in the final submissions. We list the rejected branches because they bound what the data support, and because several are natural first ideas for this benchmark. Each was registered with a fixed acceptance test before its result was seen; no failed branch was rescued by changing the test afterwards. Numbers marked † are local proxies; the others are changes in validation score. The physics-based reconstructions are in Table 11.

Table 16. Rejected branches.

ID	Idea	Evidence	Decision
State			
L0	Plain time interpolation instead of the residual model	validation −0.00867	keep model
ES5	Speed model fitted on odd March days, applied to hidden March holes	visible RMSE 2.545 → 2.006† on every corridor; validation −0.00462	rejected
	Weekday profile from the split's own masked observations	February piece 0.375 vs interpolation 0.790†	rejected
TS1	Covariance adaptation of the state model	lost on all ten panels, both development months†	rejected
TS2	Group-RMSE training loss	joint +0.00007† but S_{state} fell in both months	failed gate
TA2	Regional spatial modes	lost on all ten panels, both months†	rejected
S13	Density correction allocated by LWR edge coverage	February improved 5 of 10 panels (8 required)†	failed gate
FS1 arms	Full flow replacement; replacement only under R3; rank-24 and flow-only wider gates	lost or did not win development†	not used
	Replace short-gap speeds by the interpolant	ES1 differs from the interpolant by 1.39 km/h (mean absolute) on March one-slot cells, while the interpolant's February RMSE on such cells is 2.37 km/h†	rejected
	Restore pre-recentering speeds on hard holes (ES6)	undoes the bias shift without restoring model skill	built, withheld
	Link-by-hour bias split	RMSE 2.7560 → 2.7567† (gate −0.15)	failed gate
	December speed/flow bin calibration	January RMSE 1.6491 → 1.6450† (gate −0.05)	failed gate
	Weekday/weekend split of the link bias	1.6450 → 1.6455†	failed gate
	One-sided and long-gap shells	signal 732 vs bar 850,000 (cell-weighted km^2/h^2)†	too small
	Flow recentering on one-slot cells	March link offset 10.9 veh/h RMS (bar 20)†	too small
ES0	Structured inner masks for self-supervision	target and visible-cell gap geometry already identical; structured masks worsened the match†	closed
Onset			
U3	Training frequency prior at T+30 (top three links)	leave-one-month-out 0.558†; validation onset IoU 0.626 → 0.450	rejected
CS3	Complete onset demand history from TRAIN ramp profiles	validation onset IoU 0.750 → 0.712	rejected
RQ54	Direct onset action-value regression	training gain, April loss†	rejected
RQ60	Whitened correlation for onset retrieval	lost April and held TRAIN†	rejected
	Bottleneck placement (highest flow/capacity, largest occupancy rise)	IoU 0 on all 40 published onsets† vs analogue 0.936	rejected
	Same-month (earlier-day) onset priors	0.105 and 0.333 vs analogue 0.776 and 0.726†	rejected
	More or fewer onset neighbours, frozen on odd dates	50 neighbours 0.706 vs deployed 0.704 on even dates†	no gain
	Reranker over onset actions	0.691 vs 0.704† on even dates	rejected

continued on next page

ID	Idea	Evidence	Decision
Ongoing			
RC1	RQ7 ongoing with RQ12 onset bundle	validation −0.00798 vs J1 queues	rejected
CS2	Five-minute analogue origins	validation −0.0039	rejected
CF3	Occupancy-aware classifier RQ59	+0.0074 to +0.0085† on March; validation −0.0001	selected, lost
Q22	Neural blend with the Q19 classifier	failed its confirmation gate†	rejected
Q34	Better masked-history imputation	history RMSE 0.055 → 0.026†, forecast lost on both samples	rejected
Q36	Explicit site identifiers	failed confirmation†	rejected
TQ1, TQ2	Event-census queue models	TQ2 February +0.0038, published June −0.0010†	failed gate
FQ2, FQ3	Exact Bernoulli first-crossing law; three-model blend	April confirmation losses†	rejected
RQ49–52	Copula ranks; phase experts; space–time binary field; hindcaster	March or April losses, or no future-overlap gain†	rejected
RQ53P	Partial likelihood over 64 trajectories per link	March +0.0042, April +0.0013† (bar 0.003)	failed gate
RQ55	Better optimization of the RQ48 utility	higher objective, lower overlap in both months†	rejected
RQ56–58	Per-horizon classifiers; auxiliary thresholds; date-cluster bagging	below the 0.003 bar or losses†	rejected
RQ61–62	Scaling (similarity) augmentation of the traffic law	lost confirmation, or lost to RQ59†	rejected
	Front rollout from the legal hour	0.234 vs persistence 0.755†	rejected
	Median remaining-duration rollout	0.263 vs forward-fill 0.665†	rejected
	Yesterday's masked queue; clock-time prior	0.249 vs 0.569; 0.538 vs 0.718†	rejected
	History-only queue-dynamics model	0.831 vs analogue 0.836 (bar 0.856)†	failed gate
	Rerankers, per-link copies, neighbour-bit model	see Section 7.6	failed gates
	Same-month March library; March-fit binary model	0.324 and 0.794 vs 0.836† on even dates	rejected
	Newell envelope rolled six steps	ongoing 0.162 vs 0.866; onset 0.006 vs 0.675†	rejected
Demand			
T4-v2, v4, v5	Smooth correction, entropy fit, gravity seed	implied S_{ODME} 0.468, 0.843, 0.622 vs 0.994	rejected

Three patterns recur across these results. First, models fitted to the evaluation month's visible cells (ES5, same-month profiles, the March library) learned that month's visible behaviour and did not transfer to its hidden cells. Second, physically motivated dynamics (rollouts, envelopes, solvers) lost to empirical analogues and to temporal evidence, because the forcing that matters, namely incidents and the moment of breakdown, is not contained in the twelve history slots. Third, better local objectives (the RQ55 utility, the Q34 imputation) did not imply better forecasts; improvements must be measured on the target score, on held-out data.

11 Discussion

11.1 What the benchmark taught us about traffic

1. **At five-minute resolution, time beats space.** Links are short (many are a few hundred metres) and masked holes span one to five links. A wave crosses a link in seconds, so a link's own readings five minutes before and after carry more information about a hole than its neighbours at the same moment (Sections 5.8 and 6).
2. **Missing is not free flow.** About a quarter of last readings are missing or ineligible. Treating a dropped sensor as "not queued" cost 0.135 ongoing IoU; the forecaster must carry the last observed state forward.
3. **Onsets are recurrent in location and random in extent.** The head of a new queue is usually a known bottleneck, but how far it extends by $T{+}30$ depends on when breakdown happens inside the last five minutes, which the legal hour cannot show.
4. **Queues have memory beyond the hour.** Clearing queues in the training windows are old (300–780 minutes) and small; most growing ones are 10–20 minutes old. Queue age is visible only before $T{-}60$, outside the contract we kept.
5. **A new month brings a link-level offset.** A model trained on other months carries per-link means into the new month that the month's own anchors contradict. A label-free recentering on short gaps removed part of it.
6. **With a weak prior, demand is an optimization problem, not a modelling one.** The exact regularized solve on the split's own counts reached an implied 0.994.

11.2 Where the remaining score is

Table 17 combines the identified components with the organizers' perfect-answer scores. The state block B is bounded rather than identified after CT1 and CF1; the ongoing value inherits that bound.

Table 17. Error budget of ES4 on the validation month, in units of the composite score. "Room" is the ceiling minus our contribution; the value of B is bounded rather than identified.

Block	Component value	Contribution	Ceiling	Room
Demand, 0.20 S_{ODME}	0.9938	0.1988	0.2000	0.0012
Onset, 0.15 IoU	0.750	0.1125	0.1500	0.0375
Ongoing, 0.15 IoU	0.827–0.832	0.124–0.125	0.1500	0.025–0.026
State and physics, B		0.4358–0.4366	0.4945	0.058
Total		0.87187	0.9945	0.1226

The budget also bounds what further queue work can achieve. With our state and demand blocks unchanged, a perfect queue forecast ($S_{\mathrm{queue}} = 1$) would reach a composite score of about 0.935. Larger improvements therefore require a better state-and-physics block, most plausibly in conservation, where the block has 0.058 of room and our local instruments are weakest.

11.3 Limitations

- No private score was available for either file at the time of writing, and the validation difference between them (0.00196) is far below the queue-sampling standard deviation.
- Local proxies were reused across many studies; March and April visible events in particular were examined repeatedly, and their confidence intervals do not correct for that reuse.

- The three ensemble members share features and overlapping samples. Averaging reduces sampling variance, not bias they share.
- The gap recentering is a link mean, not a model of shocks; gaps of eight or more slots, which failed the bias test, keep the uncorrected field.
- The queue selection problem is open: the candidate set contains much better masks than the ones chosen.
- Our Task 2 contract is stricter than the rules. It excludes slot T and same-day data before $T-60$, which the rules allow, and so leaves the queue-age signal unused.
- The local physics proxy uses observed accumulation in place of the withheld boundary flux and is not the official physics score.
- The all-month classifier weights behind ES1 were not preserved, so regenerating ES4 retrains that model; whether the result is identical to the original has not yet been tested (Section 12).

11.4 Future work

- **Learn the choice among whole masks.** Train a ranker on first-crossing windows whose target is each candidate's realized IoU, and decide by expected IoU over structured outcomes rather than a fixed mixture [2]. The 0.10 gap between the chosen and the best candidate is the largest identifiable lever.
- **Model onset extent as breakdown timing.** Treat the extent at $T+30$ as a function of a breakdown time with a hazard learned from the legal hour, instead of copying neighbour extents.
- **Use legal same-day context.** The rules allow data before $T-60$. Queue age is a strong predictor of clear-versus-grow on training windows and deserves a pre-registered test.
- **Conservation as a soft target inside the learner.** Every hard physics projection lost. A penalty on the conservation residual of neighbouring predictions, learned jointly with the residual model, has not been tried.
- **Hierarchical month offsets.** Extend the recentering to flow and to longer gaps with a shrinkage estimator that borrows strength across links, instead of a hard 100-cell threshold.

12 Reproducibility

12.1 Environment and hardware

All production runs used CPU only on one AWS EC2 `r7i.2xlarge` instance (8 vCPU, 64 GiB RAM, Ubuntu with glibc 2.39) and Python 3.12.3 with the pinned packages in Table 18. No GPU is needed. The clean regeneration of TF3 took 1 h 33 min with six threads, including rebuilding every cache from the raw release. Regenerating ES4 is estimated at about 3 hours or less and both files at about 4.5 hours or less; neither total has been measured as one run. Peak memory was not recorded, so 64 GiB is what was used and not a proven minimum. The release is 7.76 GiB as a zip file, its dense caches take 4.1 GiB, and we recommend at least 40 GiB of free disk.

Kaggle Notebook. The method cannot be run inside a Kaggle Notebook as packaged and was never run in one. It was developed and operated as a chain of command-line stages under an exact pinned environment on a larger machine than a standard notebook provides. We therefore supply a repository and a written account of how it was run, `REPRODUCIBILITY.md`, which lists the versions, preprocessing steps, seeds, commands and verification status.

Table 18. Pinned inference environment (`requirements.txt`) and the licence of each dependency.

Package	Version	Licence
NumPy [18]	2.5.3	BSD-3-Clause
pandas	3.0.6	BSD-3-Clause
SciPy [17]	1.18.1	BSD-3-Clause
PyArrow	25.0.1	Apache-2.0
LightGBM [12]	4.7.0	MIT
scikit-learn	1.9.1	BSD-3-Clause

12.2 Commands

The repository has one entry point. The competition release is not redistributed; `--release-root` is the directory that contains `submission_key.csv`, `corridors/` and `task2/`.

```
# 1. verify the final archives and the TF3 code bundle (standard library only)
python reproduce.py

# 2. regenerate TF3 from the raw release and the saved weights
python -m pip install -r requirements.txt
python reproduce.py --submission tf3 --release-root /path/to/release \
                    --out /path/to/new_tf3 --threads 6

# 3. regenerate ES4 from the raw release, including EQ1 retraining
python reproduce.py --submission es4 --release-root /path/to/release \
                    --out /path/to/new_es4 --threads 6

# 4. regenerate both selected files, sharing one set of caches
python reproduce.py --submission all --release-root /path/to/release \
                    --out /path/to/new_all --threads 6
```

12.3 What has been verified

- **TF3, end to end.** The bundled driver rebuilds every cache and prediction component from the release and the saved TRAIN weights, then compares all 6,980,503 rows with the submitted file. On the clean run every queue label matched exactly and the largest continuous difference was 1.82×10^{-12}. This verifies saved-weight inference; the refit procedure, seeds and inventories are recorded, but an independent retraining was not run.
- **ES4.** The final archive is in the repository and its checksum is verified by `reproduce.py`. Row-level comparison confirms that ES4 is TF3 plus 247 ongoing-queue labels and 5,825,667 recentered speeds. `code/regenerate_es4.py` runs the original stage scripts from the release through ES1, ES2, ES3 and ES4 and compares every row with the frozen file. It has been exercised on synthetic data but **has not yet been run on the real release**. The EQ1 classifier weights existed only on the instance used during the competition, which no longer exists, so the command retrains EQ1 from the release; whether that model is identical to the original is not yet known. Only the 247 EQ1-dependent labels depend on this. TF3 scored 0.00196 (0.22 %) below ES4 on the validation month.

12.4 Artefacts and checksums

Table 19. SHA256 of the main artefacts. The first three are verified by `python reproduce.py`.

Artefact	SHA256
`submissions/submission_TF3.zip`	`2ae94d422399c7248297487fc8f45d737de0e024fbad32b6a360c290bbabca7a`
`submissions/submission_ES4.zip`	`ce03386b42e33bc6c726b09c0b64b791f294ea3abeb88d7e310552bb080c04bb`
`vendor/reproduction_TF3.zip`	`dce93b3f2a38d27014497ec1a563cc1e2d032da11c499234f226c72093678cd7`
TS3 frozen rule	`61499049f1cced66927eff4c9e9c99bb611093c33c3a31de98ebd436940883d5`
TS3 final model manifest	`8ed505b54215cd88706e95dedc5b4fbee3f435e228560f8d71d018b546cf6acb`
TS3 registered protocol	`fcc9f299f32bc0df8dc68749df21ceb1d2bc2609fc3d2a21b15a85913c84aec0`
Q46 ongoing classifier (TF3)	`2816e1d3f3c233416327591c6659cd9e80a25f0054431f1b1d4ee23735b2ab46`
EQ1 ongoing classifier, 800 trees (ES4)	`cb626813e621064594f6956344df81d7d8879e65ebc23ae2f164a2f8f6f9ac9f`

12.5 Seeds and data membership

TS3 members draw their training rows with `random_state` equal to the member seed (20261104 or 20261204) plus the panel index; the RS1 models use LightGBM seed 20261001 and samples drawn with 20261001 plus the panel index; Q46's training windows use 20261046 plus the panel index; EQ1's additional windows use 20261240 plus the panel index; both classifiers use LightGBM seed 20261002. At inference, the only seed is the S10 randomized start, 20261003 + 100× panel index + month. The fit manifests in the TF3 bundle record the fitting months, row counts and input hashes of every saved model, and two independent audits recomputed every stored weekday profile (31,783,763 fitting rows) exactly.

12.6 External data and licence declaration

No external dataset, pretrained model, paid or rate-limited API or proprietary dependency was used. The only data are the competition release, used under the competition rules and not redistributed here. All software dependencies are open source under the licences in Table 18. Figures were produced with Matplotlib and the report with LaTeX. Our own code is released under the MIT licence.

13 Conclusion

We built a pipeline for TrafficFlowBench through measured, single changes and kept a strict information boundary throughout. Its strongest parts are simple: learn the residual of temporal interpolation, choose whole onset masks by expected IoU, carry dropped sensors forward, and solve the demand system exactly. Its weakest part is equally clear: the choice among good candidate queue futures. Using the additivity of the composite score, we treated our evaluated submissions as a controlled ablation. It showed that state improvements selected on held-out months transfer to a new month, whereas queue proxies often do not. We hope that the negative results, in particular that four kinds of physics-based reconstruction lost to temporal evidence on held-out months, will be useful to future work on this benchmark.

Both final submissions are documented and checksummed, and the reconstruction submission TF3 regenerates from the raw release with a single command.

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A Evaluated submissions

Table 20. All evaluated submissions in chronological order, with their validation scores. The two final submissions are in bold. Generated from `docs/data/submissions.csv`.

#	ID	Content	Validation score
1	BASE	Organizer baselines for all tasks	0.55343
2	T4-v2	ODME: prior × smooth correction	0.47983
3	T4-v4	ODME: KL/entropy fit to the prior	0.55476
4	T4-v5	ODME: KL from a gravity seed	0.51071
5	T4-v1	ODME: exact NNLS on the split's own data	0.58501
6	V1	First GBDT state, LightGBM queues	0.80613
7	P-persq	V1 state with persistence queues	0.71192
8	P-noonset	V1 with onset rows set to zero	0.71024
9	V1-nh	V1, ongoing tree on unfiltered windows	0.80946
10	P-slotT	Ongoing set from slot T (diagnostic)	0.82669
11	U1	Legal onset classifier (12 slots)	0.80579
12	U2	U1 with forward-filled ongoing persistence	0.82600
13	J1	U2 with the joint-mask onset analogue	0.84461
14	L0	U2 with time-interpolated state	0.81733
15	U3	U2 with an onset frequency prior	0.79958
16	RC1	RQ7 ongoing and RQ12 onset	0.83663
17	RC2	RC1 with RS1 state	0.84637
18	CS1	RS1 state, RQ2 ongoing, J1 onset	0.85849
19	CS2	CS1 with origin-analogue ongoing	0.85459
20	CS3	CS2 with onset demand completion	0.84885
21	CT1	S5 speed correction, Q19 ongoing	0.85888
22	CT2	CT1 with S10/S12 state	0.86039
23	CF1	FS1 gate, Q46 ongoing	0.86488
24	CF2	CF1 with RQ48 decoder	0.86957
25	CF3	CF2 with occupancy classifier	0.86947
26	**TF3**	CF2 with three-member ensemble	**0.86991**
27	ES1	TF3 with all-month ongoing classifier	0.87030
28	ES2	ES1 with gap-1 recentering	0.87069
29	ES3	ES2 with gap-2 recentering	0.87113
30	**ES4**	ES3 with gap 3–6 recentering	**0.87187**
31	ES5	ES4 with March-fit speed model	0.86725

B Glossary of identifiers

Table 21. Identifiers used in this report.

ID	Meaning
BASE	The organizer's baselines for all four tasks, evaluated as an anchor (0.55343).
T4-vn	Task 4 probes: only demand rows differ from BASE. v1 is the exact split-specific NNLS that every later file uses.
V1	First full file: residual GBDT state, LightGBM queues reading 13 slots including T. Superseded.
P-$*$	Probes that isolate one component: P-persq (persistence queues), P-noonset (onset zeroed), P-slotT (diagnostic, reads slot T).
U1, U2, U3	First files under the twelve-slot contract: a twelve-slot onset classifier; adding forward-filled ongoing persistence; adding an onset frequency prior.
J1	U2 with the joint-mask onset analogue. Its onset and demand rows are retained in both final submissions.
L0	U2 with time-interpolated state; an ablation of the residual model.
RS1	Enhanced residual state model (92 features, density reconciliation), refit on all nine TRAIN months.
RQn, Qn	Queue research arms. RQ2: trajectory-analogue decoder. Q19: local and distance-scaled ongoing features. Q46: first-crossing classifier. RQ48: quarter-weight decoder utility. RQ59: occupancy classifier.
RCn, CSn	Candidates combining RS1 state with queue variants.
S5, S10, S12	Free-branch corrections: rank-8 covariance speed; rank-12 temporal factor flow; spatial-covariance conditional speed.
FS1	The widened, half-strength free-branch gate combining S10 and S12.
CTn, CFn	Later candidates. CT1: S5 and Q19. CT2: adds S10/S12. CF1: adds FS1 and Q46. CF2: adds RQ48. CF3: adds RQ59 (not retained).
TS3, TF3	The three-member fixed sample ensemble of the state model, and the submission that carries it (final).
EQ1	Ongoing classifier refitted on first crossings from all nine training months, 800 trees.
ES1 to ES5	TF3 with EQ1 (ES1); with gap-1, gap-2 and gap 3–6 recentering (ES2 to ES4, ES4 final); with a March-fit speed model (ES5, not retained).
Shell	Target cells whose farther temporal anchor lies a given number of five-minute slots away.
Proxy	Any locally computed score; never a validation score.